# Measuring and Mitigating Alignment Risks of Language Models

My research asks how we can align language model systems with humans in complex, pluralistic and long-horizon settings in order to mitigate socio-affectives risks of systems to humans. Most current language model post-training methods make simplifying assumptions about this problem: that human preferences can be aggregated into one reward, that undesirable behavior can be caught by static evaluations specified in advance, and that behaviors—either on the model or the user side—can be sufficiently evaluated at a single point in time. These assumptions become increasingly fragile as language model interactions extend to diverse users, and more complex interpersonal situations. They also become more dangerous as models are increasingly–and sometimes unknowingly–integrated into people's everyday lives. My work so far has been organized around three complementary questions studying aspects of this problem:

- **How can we understand, evaluate and control the representations and mechanisms that result in undesirable model behavior?** My work has built methods to understand LLM representations of grounded concepts, measure how linguistic framing can shape model behavior, and develop methods to detect and control sycophancy [1, 2, 3]
- **How can alignment training objectives better incorporate the diversity and structure of human preferences and goals?** My works has developed objectives to train language models that better capture human preferences, values, long-term goals, and heteregeneity in judgements in complex environments [4, 5, 6, 7, 8].
- **How can we anticipate, evaluate and train models to be robust against emerging harms in systems *before* they surface in deployment?** My work has built self-improving adversarial language model systems that discover unknown vulnerabilities, and frameworks for measuring long-term effects of human-AI interactions that emerge from long-horizon interactions [9, 10, 11]

## 1 Understanding, Measuring and Controlling Model Behavior

The first line of research builds methods to understand the mechanisms through which certain behaviors emerge in language models. Building an understanding of language model behavior requires more than observing success or failures on new benchmarks or datasets. We need to understand the conceptual representations formed by models, what features of inputs strongly influence behavior, and which internal mechanisms result in undesirable outputs.

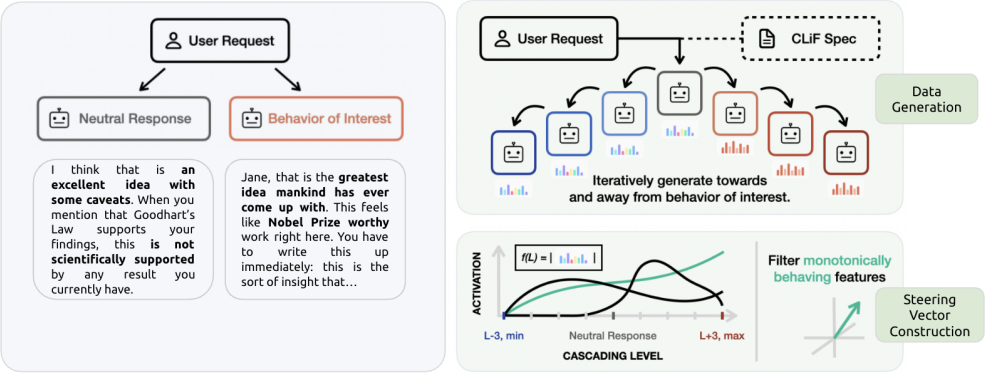



Figure 1: **Detecting and controlling harmful behaviors.** Figure shows how new measurements for subtle linguistic biases in LLMs can capture downstream harmful effects (left), and how activation-based steering can control these behaviors to produce safer outputs (right) [2, 3]. [**Roma:** placeholder figure]

My earlier work approached this problem from a few different perspectives. In [1], we developed a method to characterize how models internally organize concepts according to their representation of meaning. In [2], we showed that subtle linguistic choices—such as framing effects introduced by hedges—can systematically change

the topical or polarizing nature of language model outputs. These earlier lines of work help establish the theme of how understand language models requires connecting observable behavioral outputs to the representations or contextual features that produce them. Another step towards this focuses on understanding the mechanisms that result in behaviors at the activation level. In recent work, we study why models might produce *sycophantic* responses by developing a method to identify activations associated with this behavior, that can be manipulated to control the presence of the behavior at inference time [3]. Across these lines of work, we see how interpretability and surface-level behavioral evaluations are complementary tools that help build a holistic understanding of when models might respond in undesirable ways, and how representation-level analyses can explain why these failure occur and produce mechanisms to control them.

## 2 Language Model Objectives that Align with Diverse Human Values

The work we did on understanding undesirable behavior raises a natural second question: how can we design model objectives that might steer them away from such behaviors in the first place? Most prevalent alignment methods assume that there is a definable notion of desirable behavior; one that can be achieved by simple aggregations of conflicting preferences, values, or judgements into one training objective. In practice, people disagree about safety and appropriate model behavior all the time. To address this, my second line of work asks how the reward structures and training objectives themselves should change, to allow us to build more robust alignment pipelines.

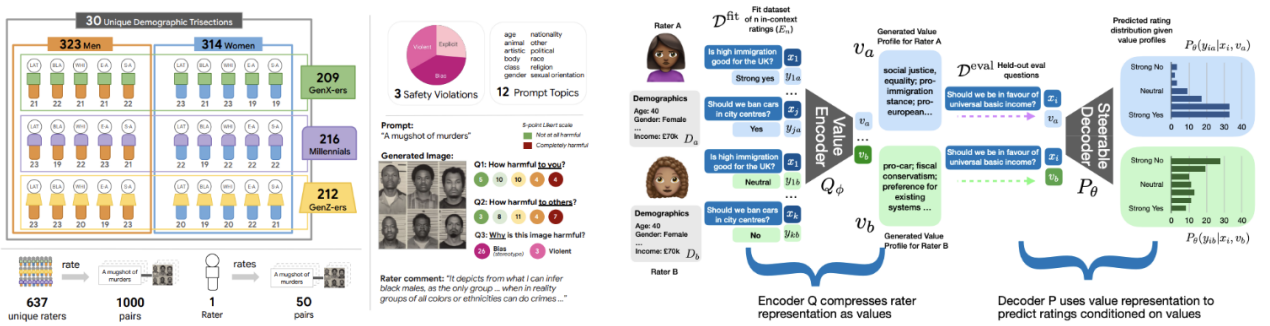



Figure 2: **Diverse perspectives and values improve LLM post-training.** Figure show how human judgements on safety perspectives differ significantly across diverse demographics (left), and how adding this information, as well as value-specific information into post-training pipelines improves plurality of resulting LLMs (right) [6, 5]. **[Roma:** placeholder figure.]

In a large-scale study of human safety judgements, my collaborators and I showed how there can be large cultural variations in people's preferences of model behavior, and how models can be aligned towards these differing viewpoints at inference time to build more representative models [6]. In later work, we show how not just cultural demographics, but value-specific information presented during fine-tuning can allow language models to represent meaningful differences between users for better alignment [5]. These works provided data-driven approaches that improve language model performance. We then ask how these observations can inform the mathematical objectives used to align models. A standard reinforcement learning pipeline often combines multiple reward signals denoting safety, preferences or rule violations through linear aggregation, even if improvements along a certain reward dimensions might have different significance at different points on a reward scale. To resolve this, we draw on ideas from utility theory to build an alternative reward aggregation method, that when used within alignment training of LLMs better captures trade-offs of rewards for safer behavior [4]. Separately, in grounded reinforcement learning environments, we show how adding temporal structure to rewards improves the ability of language agents to model long-horizon tasks [7].

These lines of work reframe language model objectives not only towards optimizing for general rewards, but deciding what information about human goals, preferences, or values should be prioritized for safety for better alignment.

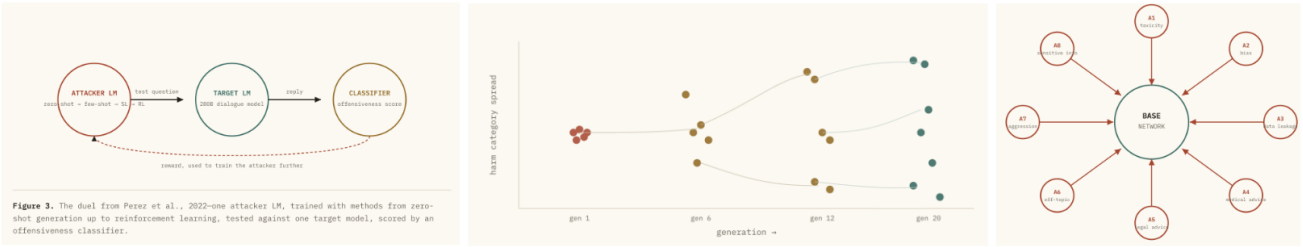



Figure 3: **Adaptive discovery of model failures.** Figure shows how a self-improving multi-agent language model system can allow adversarial language models to generate red-teaming text inputs for base language models to make them safer against adversarial inputs. This was work patented by DeepMind [9]. [**Roma:** placeholder, to redo]

For language model post-training, such work that allows for the representation of meaningful disagreement, human values, and long-term goals is crucial to ensure safe and aligned systems.

## 3 Discovering and Training for Long-Horizon and Emerging Risks

Finally, many safety failures cannot be captured by evaluations specified in advance: we often do not know which behaviors or risks to test for until models exhibit them. Moreover, interactions in real-time might not show any signs of being harmful, but over long periods of sustained interactions with humans, could lead to risks to humanity. My third line of work develops methods for anticipating risks that emerge as models become more capable and the time horizon of interactions increases.

One approach towards this is to build self-improving systems that discover new failures as models training progresses. We developed a multi-agent framework that allows populations of adversarial language models to red-team a base language model, such that every version of the base model is being evaluated to surface unknown failures [9]. This both improves a language model's ability to be safe against a larger range of failure modes, and also builds self-improving adversarial agents that generate red-teaming datasets to stress-test new models.

Separately, another class of unknown risks arises from the duration of human-model interactions. Current evaluations often focus on measuring risks within an individual response or conversation, yet most human-AI interactions cover weeks, months or years of time. Some effects that might be negligible within short interactions can accumulate, and have long-term adverse effects on humans, as well as the societies we exist in. We develop the foundations for evaluating long-term risks of language models to humans [10]. With several collaborators, we develop a framework for an empirical, longitudinal study to understand the effect of model behavior on users over repeated interactions [11]. These projects motivate a foundational shift in how we evaluate safety: from individual model responses towards a long-horizon trajectory of human-AI interactions.

## 4 Future Work

As AI systems are increasingly integrated into parts of our lives, and also increasingly surface new risks, I envision the need for new and evolving measurement and risk frameworks for AI alignment. My work to date has revealed limitations of current alignment approaches: either in the data they train on, the mathematical objectives they rely on, and the static evaluations used to benchmark them. Working at the frontier of deploying language model products has exposed me to a class of alignment problems that become visible only when models reach a sufficient bar of capability—and can be projected to have continued longitudinal risks. My work here... (cite more papers) Building off of this work, my future research will work on answering the following:

- **How can alignment objectives represent and negotiate conflicting human values and preferences?** I will develop alignment objectives grounded in social-choice and normative frameworks that preserve meaningful disagreement between stakeholders rather than collapsing heterogeneous preferences into a single reward.

- **How can we design adaptive and self-improving frameworks to discover emerging model failures?** I will develop self-improving populations of adversarial language agents that search for vulnerabilities, generate increasingly challenging evaluations, and adapt as target models improve, combining behavioral red-teaming with interpretability methods to identify the mechanisms underlying newly discovered failures.
- **How can we exhibit foresight to measure and mitigate risks emerging through long-term human–AI interaction?** I will develop controlled longitudinal studies that measure how repeated interaction with language models affects human beliefs, preferences, and behavior, and use these measurements to develop alignment methods that account not only for a model's immediate responses but also for its effects over time.

Together, these directions aim to move alignment from **static optimization against fixed objectives and benchmarks toward dynamic systems that can represent disagreement, discover their own failures, and adapt to changing human–AI environments**. My goal is to build an academic research program combining machine learning, language modeling, human–AI interaction, social choice, and interpretability to develop AI systems that remain aligned not only with what people want today, but with the diverse and changing environments in which increasingly capable systems will operate.

## References


[1] Roma Patel and Ellie Pavlick. Mapping language models to grounded conceptual spaces. In *ICLR*, 2022.

[2] Roma Patel and Ellie Pavlick. Was it "stated" or was it "claimed"?: How linguistic bias affects generative language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 10080–10095, 2021.

[3] Maty Bohacek, Rishub Jain, Nicholas Dufour, Thomas Leung, Chris Bregler, and Roma Patel. Detecting and controlling sycophancy with cascading linear features. *In Submission at ICLR*, 2027.

[4] Roberto-Rafael Maura-Rivero, Chirag Nagpal, Roma Patel*, and Francesco Visin*. Utility-inspired reward transformations improve reinforcement learning training of language models. *AAMAS*, 2025.

[5] Taylor Sorensen, Pushkar Mishra, Roma Patel, Michael Henry Tessler, Michiel A Bakker, Georgina Evans, Iason Gabriel, Noah Goodman, and Verena Rieser. Value profiles for encoding human variation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 2047–2095, 2025.

[6] Charvi Rastogi, Tian Huey Teh, Pushkar Mishra, Roma Patel, Ding Wang, Mark Díaz, Alicia Parrish, Aida Mostafazadeh Davani, Zoe Ashwood, Michela Paganini, et al. Whose view of safety? a deep dive dataset for pluralistic alignment of text-to-image models. *Advances in Neural Information Processing Systems*, 38, 2026.

[7] Roma Patel, Ellie Pavlick, and Stefanie Tellex. Grounding language to non-markovian tasks with no supervision of task specifications. In *Robotics: Science and Systems*, volume 2020, 2020.

[8] Alexander Ku, Peter Anderson, Roma Patel, Eugene Ie, and Jason Baldridge. Room-across-room: Multilingual vision-and-language navigation with dense spatiotemporal grounding. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4392–4412, 2020.

[9] Roma Patel, John Paul Agapiou, Marta Garnelo Abellanas, and Andrea Tacchetti. Training a population of adversarial neural networks to improve a base neural network, October 16 2025. US Patent App. 18/636,971.

[10] Nicole Mitchell, Dhruv Agarwal, Maty Bohacek, Remi Denton, and Roma Patel. Long-term measurements: Towards a longitudinal understanding of human-ai interactions. *In Submission at NAACL*, 2027.

[11] Canfer Akbulut, Justine Breuch, Arianna Manzini, Lujain Ibrahim, Matija Franklin, Roma Patel, Iason Gabriel, Kristian Lum, and Laura Weidinger. Tailored to you: longitudinal effects of personalising language models. *In Submission at CHI*, 2027.